AMC 12A 2024 Problem 21 Analysis

Comprehensive Strategic Framework Application

Problem Analysis Framework v2.1

October 28, 2025

Problem Statement

Suppose that $a_1 = 2$ and the sequence (a_n) satisfies the recurrence relation

$$\frac{a_n - 1}{n - 1} = \frac{a_{n-1} + 1}{(n-1) + 1}$$

for all $n \geq 2$. What is the greatest integer less than or equal to

$$\sum_{n=1}^{100} a_n^2?$$

(A) 338,550 **(B)** 338,551 **(C)** 338,552 **(D)** 338,553 **(E)** 338,554

1 Problem Configuration Analysis

Problem Configuration Analysis

A. Problem Classification

Problem Type: To Find (computational problem with sequences and series)
Difficulty Band: Late-band (Problem 21 of 25)

- Expected time: 5-7 minutes
- Difficulty level: High (84% position)
- Requires recurrence relation solution
- Involves summation and estimation

Competition Context: AMC 12A (75 minutes, 25 problems, multiple choice)
Mathematical Domain:

- Primary: Sequences and Series (recurrence relations)

- Secondary: Algebra (manipulation and simplification)
- Tertiary: Summation Formulas (sum of squares, telescoping)
- Quaternary: Estimation (bounding infinite series)

B. Problem Structure Identification

Given Information:

- Initial condition: $a_1 = 2$
- Recurrence: $\frac{a_n - 1}{n-1} = \frac{a_{n-1}+1}{n}$ for $n \geq 2$
- Goal: Find $\lfloor \sum_{n=1}^{100} a_n^2 \rfloor$
- Answer choices differ by only 1

Constraints:

- Must solve recurrence to find explicit formula
- Must evaluate sum to 100 terms
- Need floor function (greatest integer)
- Answer choices very close together (requires precision)

Objective: Find the greatest integer $\leq \sum_{n=1}^{100} a_n^2$.
Hidden Assumptions:

- The recurrence can be solved explicitly
- The sum will not be an integer (otherwise floor irrelevant)
- Small terms matter (answer choices differ by 1)
- Estimation techniques needed for $\sum \frac{1}{n^2}$

Answer Choice Leverage:

- All choices in narrow range: 338,550-338,554
- Sequential: differ by exactly 1
- Large values suggest sum dominated by large terms
- Middle choice (C) might be trap if calculation error
- Need precise calculation to distinguish

C. Strategic Orientation

Hypothesis: The recurrence transforms into a simpler form via substitution.
Conclusion: Solve for a_n, compute a_n^2, sum, and apply floor function.
Notation:

- Let $b_n = n(a_n - 1)$ or $b_n = na_n$ (clever substitution)
- Use standard summation formulas
- Denote $S = \sum_{n=1}^{100} a_n^2$

Intuitive Approach:

- Transform recurrence via substitution
- Use telescoping to find explicit formula
- Recognize $a_n = n + \frac{1}{n}$ pattern
- Split sum into manageable parts
- Estimate $\sum \frac{1}{n^2}$ term

Cross-Domain Potential:

- Algebraic: Manipulation and substitution
- Analysis: Series estimation and bounds
- Number theory: Floor function application

Key Insight

Critical Discovery: The recurrence simplifies beautifully via substitution.
Method 1 - Substitution $b_n = n(a_n - 1)$:
Starting from: $\frac{a_n - 1}{n-1} = \frac{a_{n-1}+1}{n}$
Cross-multiply: $n(a_n - 1) = (n-1)(a_{n-1} + 1)$
Let $b_n = n(a_n - 1)$, then:

$$b_n = (n-1)(a_{n-1} - 1) + 2(n-1) = b_{n-1} + 2(n-1)$$

This telescopes to: $b_n = b_1 + 2(1 + 2 + \cdots + (n-1)) = 1 + n(n-1) = n^2 - n + 1$
Therefore: $a_n = \frac{b_n + n}{n} = \frac{n^2+1}{n} = n + \frac{1}{n}$
Method 2 - Substitution $b_n = na_n$:
Transform to: $na_n - (n-1)a_{n-1} = 2n - 1$
Let $b_n = na_n$, then: $b_n - b_{n-1} = 2n - 1$
Telescoping: $b_n = b_1 + \sum_{k=2}^{n}(2k-1) = 2 + (3 + 5 + \cdots + (2n-1)) = 1 + n^2$
Therefore: $a_n = \frac{1+n^2}{n} = n + \frac{1}{n}$

Key Result: $a_n = n + \frac{1}{n}$, so $a_n^2 = n^2 + 2 + \frac{1}{n^2}$

2 Strategic Framework Application

Strategic Framework Application

A. Psychological Strategies

Mental Toughness:

- Problem 21 requires algebraic persistence
- Multiple steps: solve recurrence, sum, estimate
- Don't panic at complex recurrence
- Trust that clever substitution exists

Creativity Cultivation:

- Think: What substitution simplifies this?
- Consider: Multiply both sides by something useful
- Pattern recognition: Look for telescoping
- Guess-and-check: Compute first few terms

Iterative Refinement:

- First: Try to isolate a_n
- Second: If messy, try substitution
- Third: Look for pattern in small cases
- Fourth: Verify formula works

B. Investigation Strategies

Startup Orientation:

- Understand: This is a recurrence relation problem
- Identify: Need explicit formula for a_n
- Recognize: Substitution likely needed
- Note: Answer choices very close (need precision)

Get Your Hands Dirty:

- Compute: $a_1 = 2$
- Compute: $a_2 = \frac{5}{2}$ from recurrence
- Compute: $a_3 = \frac{10}{3}$, $a_4 = \frac{17}{4}$
- Pattern: $a_n = \frac{n^2+1}{n}$ becomes evident!

Penultimate Step:

- Final step: Apply floor function to sum
- Before that: Estimate $\sum \frac{1}{n^2}$ term
- Before that: Sum n^2 and constant terms
- Before that: Find $a_n^2 = n^2 + 2 + \frac{1}{n^2}$
- Before that: Solve for $a_n = n + \frac{1}{n}$

Wishful Thinking:

- Wish: If only recurrence was simple $a_n = a_{n-1} + c$
- Reality: More complex, but substitution helps
- Wish: If only $\sum \frac{1}{n^2}$ was known exactly
- Reality: Need to bound between 1 and 2

Make It Easier:

- Simplify: Cross-multiply recurrence first
- Simplify: Use substitution to linearize
- Simplify: Recognize telescoping pattern
- Simplify: Split sum into three manageable parts

Conjecture via Small Cases:

- $a_1 = 2 = 1 + 1$
- $a_2 = \frac{5}{2} = 2 + \frac{1}{2}$
- $a_3 = \frac{10}{3} = 3 + \frac{1}{3}$
- $a_4 = \frac{17}{4} = 4 + \frac{1}{4}$

- Pattern: $a_n = n + \frac{1}{n}$ ✓

Visualization:

- Visualize: Telescoping sum as cancellation
- Think: $\sum \frac{1}{n^2}$ converges to $\frac{\pi^2}{6} \approx 1.645$
- Understand: Why answer is 338,551 not 338,552

C. Late-Band Strategic Playbook

Answer-Choice Leverage:

- Sequential answers: 338,550 through 338,554
- Difference of 1: Small fractional part matters
- Pattern: $338,550 + k$ where $k \in \{0, 1, 2, 3, 4\}$
- Strategy: Once we get 338,550 + (1 to 2), answer is (B)

Make It Easier Application:

- Replace: Complex recurrence with explicit formula
- Replace: Sum of a_n^2 with sum of simpler terms
- Replace: Exact calculation of $\sum \frac{1}{n^2}$ with bounds
- Build: Use known formula for $\sum n^2$

Penultimate Step Focus:

- Working backwards: Need $\lfloor S \rfloor = 338,551$
- Therefore: Need $338,551 < S < 338,552$
- Therefore: Need $338,550 + 1 < S < 338,550 + 2$
- Check: $\sum \frac{1}{n^2} \in (1, 2)$ gives this ✓

D. Argument Construction Strategy

Direct Approach:

1. Cross-multiply the recurrence relation
2. Apply clever substitution to linearize
3. Use telescoping to find explicit formula

4. Square a_n and split into three sums
5. Evaluate using standard formulas and bounds
6. Apply floor function

Key Lemma: $1 < \sum_{n=1}^{100} \frac{1}{n^2} < 2$
Proof: Lower bound is trivial ($\frac{1}{1^2} = 1$). For upper bound, use comparison:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < 2$$

or group terms and bound by geometric series.

Warning

Common Mistakes:

1. **Algebraic errors in recurrence manipulation:** Losing track of signs or coefficients
2. **Forgetting initial condition:** Not verifying $a_1 = 2$ works with formula
3. **Wrong summation formula:** Using $\frac{n(n+1)}{2}$ instead of $\frac{n(n+1)(2n+1)}{6}$
4. **Misapplying floor function:** Taking floor too early or incorrectly
5. **Poor estimation:** Not bounding $\sum \frac{1}{n^2}$ correctly
6. **Arithmetic errors:** Large numbers make calculation mistakes likely

3 Tactical Methodology Selection

Tactical Methodology Selection

A. Core Mathematical Tactics

Substitution:

- Applied: Transform a_n to b_n for simpler recurrence
- Two approaches: $b_n = n(a_n - 1)$ or $b_n = na_n$
- Goal: Linearize the recurrence
- Success: Creates telescoping sum

Telescoping:

- Recognition: $b_n - b_{n-1} = f(n)$ pattern
- Application: Sum both sides from 2 to n
- Result: Most terms cancel, leaving simple expression
- Efficiency: Avoids iterating 100 times

Pattern Recognition:

- Strategy: Compute first few terms
- Observation: $a_n = \frac{n^2+1}{n}$
- Verification: Check formula satisfies recurrence
- Advantage: Direct path if substitution unclear

Estimation:

- Target: $\sum_{n=1}^{100} \frac{1}{n^2}$
- Method: Bound between simple values
- Lower: Clearly > 1
- Upper: Use $< \frac{\pi^2}{6} < 2$ or telescoping comparison

B. Domain-Specific Tactics

Sequences and Series Tactics:

- **Recurrence Solving:** Substitution, characteristic equations
- **Telescoping Sums:** Key for finding explicit formulas
- **Standard Formulas:** $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$
- **Series Bounds:** Comparison tests, grouping methods

Algebra Tactics:

- **Cross-Multiplication:** Clear fractions
- **Expansion:** $(n + \frac{1}{n})^2 = n^2 + 2 + \frac{1}{n^2}$
- **Factoring:** Recognize patterns

Estimation Tactics:

- **Bounding Series:** Upper and lower bounds
- **Known Constants:** $\frac{\pi^2}{6} \approx 1.645$
- **Comparison:** $\frac{1}{n^2} < \frac{1}{n(n-1)}$ for $n \geq 2$

C. Crossover Techniques

Algebra ↔ Sequences:

- Translate: Recurrence relation to algebraic equation
- Manipulate: Using standard algebraic techniques
- Return: To sequence interpretation

Discrete ↔ Analysis:

- Discrete: Finite sum to 100 terms
- Analysis: Use infinite series bounds
- Connection: Finite sum < infinite sum

Pro Tip

Standard Formula Quick Reference:
Always have these memorized for AMC 12:

$$\sum_{n=1}^{N} n = \frac{N(N+1)}{2}$$
$$\sum_{n=1}^{N} n^2 = \frac{N(N+1)(2N+1)}{6}$$
$$\sum_{n=1}^{N} n^3 = \left(\frac{N(N+1)}{2}\right)^2$$
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.645$$

Quick Calculation for $N = 100$:

$$\sum_{n=1}^{100} n^2 = \frac{100 \cdot 101 \cdot 201}{6} = \frac{2{,}030{,}100}{6} = 338{,}350$$

Bounding $\sum \frac{1}{n^2}$:
Lower bound: Obviously > 1 (first term alone)
Upper bound (telescoping method):

$$\frac{1}{n^2} < \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$$

$$\sum_{n=2}^{100} \frac{1}{n^2} < \sum_{n=2}^{100} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{100} < 1$$

$$\sum_{n=1}^{100} \frac{1}{n^2} < 1 + 1 = 2$$

4 Conversion Techniques Application

Conversion Techniques Application

A. Recognition Index

Primary Technique: Substitution to Linearize

- Recognition trigger: Non-linear recurrence
- Strategy: Find substitution that creates linear form
- Candidates: $b_n = n(a_n - 1)$ or $b_n = na_n$
- Result: Telescoping sum emerges

Secondary Technique: Pattern Recognition

- Recognition trigger: Small cases compute easily
- Strategy: Calculate a_1, a_2, a_3, a_4
- Pattern: $a_n = n + \frac{1}{n}$ becomes obvious
- Advantage: Bypasses substitution if pattern seen quickly

Tertiary Technique: Series Decomposition

- Recognition trigger: Sum of a_n^2 with known form
- Strategy: Split into three separate sums
- Application: $\sum(n^2 + 2 + \frac{1}{n^2})$ separates cleanly
- Benefit: Each sum uses different technique

B. Technique Selection

Why Substitution Method?

- Systematic: Works even without pattern recognition
- Rigorous: Proves formula rather than guessing
- Generalizable: Technique applies to similar problems
- Elegant: Creates beautiful telescoping

Why Pattern Recognition as Alternative?

- Fast: Immediate if pattern obvious
- Intuitive: Natural for visual thinkers
- Competition-friendly: Saves time under pressure
- Risk: Must verify formula works

Why Series Estimation?

- Necessary: Can't compute $\sum \frac{1}{n^2}$ exactly easily
- Efficient: Bounds suffice for floor function
- Standard: Common competition technique
- Precise: Gives tight enough bounds

C. Integration Strategy

Phase 1: Solve Recurrence

- Cross-multiply original recurrence
- Apply substitution $b_n = n(a_n - 1)$ or $b_n = na_n$
- Use telescoping to find b_n
- Extract $a_n = n + \frac{1}{n}$

Phase 2: Square and Decompose

- Calculate $a_n^2 = n^2 + 2 + \frac{1}{n^2}$
- Write sum: $\sum a_n^2 = \sum n^2 + \sum 2 + \sum \frac{1}{n^2}$
- Recognize standard formulas apply

Phase 3: Evaluate and Estimate

- Compute $\sum_{n=1}^{100} n^2 = 338,350$
- Compute $\sum_{n=1}^{100} 2 = 200$
- Estimate $1 < \sum_{n=1}^{100} \frac{1}{n^2} < 2$
- Sum: $338,550 + (1 \text{ to } 2)$

Phase 4: Apply Floor Function

- Range: $S \in (338,551, 338,552)$
- Floor: $\lfloor S \rfloor = 338,551$
- Answer: (B)

D. Conflict Resolution

Potential Conflict 1: Which Substitution?

- Conflict: $b_n = n(a_n - 1)$ vs. $b_n = na_n$
- Resolution: Both work! Choose whichever you see first
- Lesson: Multiple valid approaches exist

Potential Conflict 2: Exact vs. Approximate

- Conflict: Need exact value of $\sum \frac{1}{n^2}$?
- Resolution: No! Bounds suffice for floor function
- Lesson: Don't overwork the problem

Potential Conflict 3: Answer Choice Ambiguity

- Conflict: Sum could be near 338,551.5
- Resolution: Better bounds show it's in (338,551, 338,552)
- Lesson: Precision matters when choices are close

5 Verification Strategy Design

Verification Strategy Design

A. Verification Level Selection

Selected Level: Level 5 - Targeted Deep Check (2-3 minutes)
Decision Tree Analysis:

- Problem difficulty: Late-band (P21) → High stakes
- Personal confidence: Moderate (many calculation steps)
- Time remaining: Assume 12-15 minutes left → Can afford thorough check
- Answer type: Close multiple choices → Need precision
- Recommendation: Level 5 (Targeted Deep Check)

B. Specialized Verification Methods

Formula Verification:
Check 1: Does $a_n = n + \frac{1}{n}$ satisfy initial condition?

$$a_1 = 1 + \frac{1}{1} = 2\checkmark$$

Check 2: Does formula satisfy recurrence?

$$\frac{a_n - 1}{n-1} = \frac{n + \frac{1}{n} - 1}{n-1} = \frac{n - 1 + \frac{1}{n}}{n-1} = 1 + \frac{1}{n(n-1)}$$
$$\frac{a_{n-1} + 1}{n} = \frac{(n-1) + \frac{1}{n-1} + 1}{n} = \frac{n + \frac{1}{n-1}}{n} = 1 + \frac{1}{n(n-1)}\checkmark$$

Check 3: Verify pattern with computed values

$$a_2 = 2 + \frac{1}{2} = \frac{5}{2}\checkmark$$
$$a_3 = 3 + \frac{1}{3} = \frac{10}{3}\checkmark$$
$$a_4 = 4 + \frac{1}{4} = \frac{17}{4}\checkmark$$

Summation Verification:
Check 4: Sum of Squares Formula

$$\begin{aligned}\sum_{n=1}^{100} n^2 &= \frac{100 \cdot 101 \cdot 201}{6} \\ &= \frac{2{,}030{,}100}{6} \\ &= 338{,}350\checkmark\end{aligned}$$

Check 5: Constant Sum

$$\sum_{n=1}^{100} 2 = 2 \cdot 100 = 200\checkmark$$

Check 6: Bounds on $\sum \frac{1}{n^2}$
Lower bound:

$$\sum_{n=1}^{100} \frac{1}{n^2} \geq \frac{1}{1^2} = 1\checkmark$$

Upper bound (telescoping):

$$\begin{aligned}
\sum_{n=1}^{100} \frac{1}{n^2} &= 1 + \sum_{n=2}^{100} \frac{1}{n^2} \\
&< 1 + \sum_{n=2}^{100} \frac{1}{n(n-1)} \\
&= 1 + \sum_{n=2}^{100} \left(\frac{1}{n-1} - \frac{1}{n} \right) \\
&= 1 + \left(1 - \frac{1}{100} \right) \\
&< 2\checkmark
\end{aligned}$$

Check 7: Total Sum Range

$$\begin{aligned}
\sum a_n^2 &= 338{,}350 + 200 + \sum \frac{1}{n^2} \\
&= 338{,}550 + \sum \frac{1}{n^2} \\
&\in (338{,}550 + 1, 338{,}550 + 2) \\
&= (338{,}551, 338{,}552)\checkmark
\end{aligned}$$

Check 8: Floor Function Application

$$\lfloor S \rfloor = 338{,}551 \text{ when } S \in (338{,}551, 338{,}552)\checkmark$$

Answer Choice Validation:
Check 9: Does Answer Match?

- Our answer: 338,551
- Choice (B): 338,551 ✓

Check 10: Why Not Adjacent Choices?

- (A) 338,550: Would need $\sum \frac{1}{n^2} < 1$ (impossible)
- (C) 338,552: Would need $\sum \frac{1}{n^2} \geq 2$ (false)
- Both ruled out ✓

C. Confidence Calibration

Pre-Verification Confidence: 80%

- Strong: Clear derivation path
- Strong: Formula checks out for small cases
- Concern: Arithmetic errors in large numbers
- Concern: Bounds on series might be wrong

Post-Verification Confidence: 97%

- Confirmed: Formula satisfies recurrence
- Confirmed: All arithmetic correct
- Confirmed: Bounds are tight
- Confirmed: Answer matches choice
- Remaining 3%: Minimal risk of oversight

Final Decision: SELECT ANSWER (B)

6 Solution Roadmap

Solution Roadmap

Complete Solution Path

Phase 1: Solve the Recurrence (2-3 minutes)
Step 1.1: Cross-Multiply the Recurrence
Given: $\frac{a_n-1}{n-1} = \frac{a_{n-1}+1}{n}$
Cross-multiply:

$$n(a_n - 1) = (n-1)(a_{n-1} + 1)$$

Step 1.2: Apply Substitution
Let $b_n = n(a_n - 1)$. Then:

$$\begin{aligned} b_n &= (n-1)(a_{n-1}+1) \\ &= (n-1)(a_{n-1}-1) + 2(n-1) \\ &= b_{n-1} + 2(n-1) \end{aligned}$$

This is a simple recurrence: $b_n - b_{n-1} = 2(n-1)$
Step 1.3: Use Telescoping

Summing from $n = 2$ to $n = N$:

$$\begin{aligned} b_N - b_1 &= \sum_{k=2}^{N} 2(k-1) \\ &= 2\sum_{k=1}^{N-1} k \\ &= 2 \cdot \frac{(N-1)N}{2} \\ &= N(N-1) \end{aligned}$$

Since $b_1 = 1(a_1 - 1) = 1(2-1) = 1$:

$$b_N = 1 + N(N-1) = N^2 - N + 1$$

Step 1.4: Extract a_n

$$\begin{aligned} n(a_n - 1) &= n^2 - n + 1 \\ a_n - 1 &= n - 1 + \frac{1}{n} \\ a_n &= n + \frac{1}{n} \end{aligned}$$

Checkpoint 1: Have I correctly solved for a_n?
Phase 2: Square and Decompose (1 minute)
Step 2.1: Calculate a_n^2

$$\begin{aligned} a_n^2 &= \left(n + \frac{1}{n}\right)^2 \\ &= n^2 + 2 \cdot n \cdot \frac{1}{n} + \frac{1}{n^2} \\ &= n^2 + 2 + \frac{1}{n^2} \end{aligned}$$

Step 2.2: Write Sum as Three Parts

$$\begin{aligned} \sum_{n=1}^{100} a_n^2 &= \sum_{n=1}^{100} \left(n^2 + 2 + \frac{1}{n^2}\right) \\ &= \sum_{n=1}^{100} n^2 + \sum_{n=1}^{100} 2 + \sum_{n=1}^{100} \frac{1}{n^2} \end{aligned}$$

Checkpoint 2: Have I correctly expanded a_n^2 and separated the sum?
Phase 3: Evaluate Each Sum (2-3 minutes)

Step 3.1: Sum of Squares

$$\begin{aligned}\sum_{n=1}^{100} n^2 &= \frac{100 \cdot 101 \cdot 201}{6} \\ &= \frac{100 \cdot 101 \cdot 201}{6} \\ &= \frac{2{,}030{,}100}{6} \\ &= 338{,}350\end{aligned}$$

Step 3.2: Sum of Constants

$$\sum_{n=1}^{100} 2 = 2 \cdot 100 = 200$$

Step 3.3: Bound Sum of Reciprocal Squares
Lower bound (trivial):

$$\sum_{n=1}^{100} \frac{1}{n^2} > \frac{1}{1^2} = 1$$

Upper bound (telescoping comparison):

$$\begin{aligned}\sum_{n=1}^{100} \frac{1}{n^2} &= 1 + \sum_{n=2}^{100} \frac{1}{n^2} \\ &< 1 + \sum_{n=2}^{100} \frac{1}{n(n-1)} \quad \left(\text{since } \frac{1}{n^2} < \frac{1}{n(n-1)} \text{ for } n \geq 2\right) \\ &= 1 + \sum_{n=2}^{100} \left(\frac{1}{n-1} - \frac{1}{n}\right) \\ &= 1 + \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots + \frac{1}{99} - \frac{1}{100}\right) \\ &= 1 + \left(1 - \frac{1}{100}\right) \\ &= 2 - \frac{1}{100} < 2\end{aligned}$$

Therefore: $1 < \sum_{n=1}^{100} \frac{1}{n^2} < 2$
Checkpoint 3: Have I correctly bounded the reciprocal sum?
Phase 4: Final Calculation (30 seconds)

Step 4.1: Sum All Parts

$$\sum_{n=1}^{100} a_n^2 = 338{,}350 + 200 + \sum_{n=1}^{100} \frac{1}{n^2}$$
$$= 338{,}550 + \sum_{n=1}^{100} \frac{1}{n^2}$$

Since $1 < \sum_{n=1}^{100} \frac{1}{n^2} < 2$:

$$338{,}551 < \sum_{n=1}^{100} a_n^2 < 338{,}552$$

Step 4.2: Apply Floor Function

$$\lfloor \sum_{n=1}^{100} a_n^2 \rfloor = 338{,}551$$

Step 4.3: Match to Answer

- Our answer: 338,551
- Choice: **(B)** 338,551

Checkpoint 4: Is my final answer correct?

Alternative Approaches

Alternative 1: Pattern Recognition Only

- Compute $a_1 = 2$, $a_2 = \frac{5}{2}$, $a_3 = \frac{10}{3}$, $a_4 = \frac{17}{4}$
- Recognize pattern: $a_n = \frac{n^2+1}{n}$
- Verify formula works for computed values
- Proceed directly to squaring and summing
- Faster if pattern seen quickly

Alternative 2: Different Substitution

- Use $b_n = na_n$ instead of $b_n = n(a_n - 1)$
- Transform: $na_n - (n-1)a_{n-1} = 2n - 1$
- Telescoping: $b_n = b_1 + \sum_{k=2}^{n}(2k-1) = 2 + (3 + 5 + \cdots + (2n-1))$
- Result: $b_n = 1 + n^2$, so $a_n = \frac{1+n^2}{n} = n + \frac{1}{n}$

- Same result, different path

Alternative 3: Use Known Value of $\frac{\pi^2}{6}$

- If you know $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \approx 1.6449$
- Conclude $\sum_{n=1}^{100} \frac{1}{n^2} \approx 1.635$ (slightly less)
- Directly get $338{,}550 + 1.635 \approx 338{,}551.635$
- Floor gives 338,551
- Faster if constant memorized

Common Pitfalls

Pitfall 1: Algebraic Errors in Substitution

- Error: Incorrectly expanding $(n-1)(a_{n-1}+1)$
- Why wrong: Sign errors or missing terms
- Prevention: Carefully expand and check each step

Pitfall 2: Wrong Summation Formula

- Common: Using $\sum n^2 = \frac{n(n+1)}{2}$ (wrong!)
- Correct: $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$
- Prevention: Memorize standard formulas

Pitfall 3: Poor Bounds on $\sum \frac{1}{n^2}$

- Error: Not bounding tightly enough
- Why wrong: Might not distinguish between answer choices
- Prevention: Use telescoping comparison to get < 2

Pitfall 4: Arithmetic Errors

- Common: $\frac{2{,}030{,}100}{6} = 338{,}350$ miscalculated
- Prevention: Double-check division with large numbers
- Tip: $\frac{100 \cdot 101 \cdot 201}{6} = \frac{100 \cdot 101 \cdot 201}{6}$

Pitfall 5: Floor Function Confusion

- Error: Rounding instead of flooring
- Why wrong: Floor always rounds down
- Example: $\lfloor 338{,}551.9 \rfloor = 338{,}551$, not 338,552

7 Competition Strategy Integration

Competition Strategy Integration

A. Time Management

Time Budget for Problem 21:

- Total allocated: 5-7 minutes (late-band problem)
- Phase 1 (Solve recurrence): 2-3 minutes
- Phase 2 (Square/decompose): 1 minute
- Phase 3 (Evaluate sums): 2-3 minutes
- Phase 4 (Final calculation): 30 seconds
- Phase 5 (Verification): 1-2 minutes

Time Checkpoints:

- After 3 minutes: Should have $a_n = n + \frac{1}{n}$
- After 5 minutes: Should have sum decomposed and evaluated
- After 7 minutes: Should have final answer
- After 8 minutes: If not done, make educated guess

B. Strategic Positioning

When to Attempt:

- Strong algebra: Attempt after P1-20
- Weak sequences: Consider skipping for P22-25
- If >12 minutes remaining: Attempt
- If familiar with recurrences: High success probability

Problem Sequencing:

- This is P21 - beginning of late band
- More accessible than P22-25 for some students
- Consider: Algebraic vs. geometric/combinatorial preference
- Strategy: Attempt if algebra is strength

C. Risk Assessment

Success Probability:

- With strong algebra: 70-80%
- With recurrence experience: 75-85%
- Standard preparation: 50-60%
- If pattern seen quickly: 80-90%

Risk Level: MEDIUM

- Difficulty: High (P21/25)
- Complexity: Multiple steps but systematic
- Time cost: 5-7 minutes
- Payoff: Same 6 points as easier problems
- Advantage: More systematic than P23-24

Risk Mitigation:

- Quick check: Can I compute a_2, a_3? (If yes: proceed)
- Quick check: Do I see pattern in small cases? (If yes: fast path)
- Quick check: Do I know $\sum n^2$ formula? (If no: risky)
- Backup: Answer choices differ by 1, so precision matters

D. Abandonment Criteria

When to Abandon:
Scenario 1: After 3 minutes, can't solve recurrence

- Action: Try pattern recognition with small cases
- If still stuck: Skip to next problem
- Reason: Fundamental blockage unlikely to resolve

Scenario 2: After 6 minutes, arithmetic errors mounting

- Action: Use answer choice structure to guess
- Reasoning: Middle choices (B,C,D) most likely

- Best guess: (B) based on bound analysis

Scenario 3: Less than 10 minutes remaining in exam

- Action: Strategic guess immediately
- Reason: Not enough time for full solution

Strategic Guessing (if must abandon):

- Observe: Answers are 338,550 through 338,554
- Pattern: Sequential, differ by 1
- Key insight: $\sum \frac{1}{n^2}$ is between 1 and 2
- Therefore: Answer is 338,550 + 1 = 338,551 or 338,550 + 2 = 338,552
- Best guess: (B) 338,551 (sum is closer to 1.6 than to 2)

E. Answer Recording

Recording Protocol:

- Write answer clearly: B
- Double-check bubble: (B) is filled
- Cross-reference: Problem 21 = bubble 21
- Watch for: Large numbers, easy to misread

F. Post-Problem Strategy

If Solved Successfully:

- Confidence: Algebra skills validated
- Time check: How much remaining?
- Next: Tackle remaining late-band problems

If Abandoned:

- No regret: P21 requires specific techniques
- Focus: Maximize points on achievable problems
- Learn: Review telescoping and substitution methods later

8 Summary and Key Takeaways

Key Insight

Critical Insights for This Problem:

1. The recurrence relation simplifies beautifully via substitution. Either $b_n = n(a_n - 1)$ or $b_n = na_n$ works, creating a telescoping sum.
2. The explicit formula $a_n = n + \frac{1}{n}$ can be found either by solving the recurrence or by computing small cases and recognizing the pattern.
3. Squaring gives $a_n^2 = n^2 + 2 + \frac{1}{n^2}$, which decomposes into three manageable sums.
4. The key to getting the right answer is bounding $\sum_{n=1}^{100} \frac{1}{n^2}$ correctly between 1 and 2. The telescoping comparison $\frac{1}{n^2} < \frac{1}{n(n-1)}$ is the elegant method.
5. With $338,550 + (1 \text{ to } 2)$, the floor gives $338,551$. Answer choices differ by 1, so precision is essential.

Pro Tip

General Strategy for Recurrence Problems:
When faced with recurrence relations:

1. **Try small cases first:** Compute a_1, a_2, a_3, a_4 and look for patterns. Often the pattern is obvious and you can guess-and-verify.
2. **Look for simplifying substitutions:** If pattern unclear, try:
   - Multiplying/dividing by n
   - Adding/subtracting related terms
   - Creating combinations like $b_n = n \cdot a_n$ or $b_n = a_n - f(n)$
3. **Aim for telescoping:** The ideal transformed recurrence has form $b_n - b_{n-1} = f(n)$, which telescopes cleanly.
4. **Verify your formula:** Always check that your explicit formula satisfies both the recurrence and initial condition.
5. **Use standard formulas:** Memorize $\sum n$, $\sum n^2$, $\sum n^3$ and know how to bound simple series.

For this problem, recognizing that $b_n = n(a_n - 1)$ or $b_n = na_n$ creates a linear recurrence is the breakthrough that unlocks everything.

9 Final Answer

$$\boxed{\textbf{Answer: (B)} \quad 338{,}551}$$

Solution Summary:

- Transform recurrence via substitution: $b_n = n(a_n - 1)$ gives $b_n - b_{n-1} = 2(n-1)$
- Telescope to find: $b_n = n^2 - n + 1$
- Extract explicit formula: $a_n = n + \frac{1}{n}$
- Square: $a_n^2 = n^2 + 2 + \frac{1}{n^2}$
- Sum parts:

$$\begin{aligned}\sum_{n=1}^{100} a_n^2 &= \sum_{n=1}^{100} n^2 + \sum_{n=1}^{100} 2 + \sum_{n=1}^{100} \frac{1}{n^2} \\ &= 338{,}350 + 200 + (1 \text{ to } 2) \\ &\in (338{,}551, 338{,}552)\end{aligned}$$

- Apply floor: $\lfloor S \rfloor = 338{,}551$